

Indian Institute of Technology Guwahati
Statistical Inference and Multivariate Analysis (MA324)
Problem Set : Linear Regression

1. Consider the simple linear regression model $y = \beta_0 + \beta_1 x + \varepsilon$ with usual assumptions on ε . Show that

$$E(MS_R) = \sigma^2 + \beta_1^2 S_{xx} \quad \text{and} \quad E(MS_{Res}) = \sigma^2.$$

2. Suppose that we have fit a simple linear regression model $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1$, but the response is affected by a second variable x_2 such that the true regression is $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$. Is the least square estimator $\hat{\beta}_1$ in the simple linear regression model unbiased?
3. Consider the simple linear regression model $y = \beta_0 + \beta_1 x + \varepsilon$, where ε 's are independent and identically $N(0, \sigma^2)$ random variables. Find the MLEs of β_0 , β_1 , and σ^2 . Is the MLE of σ^2 UE?
4. Suppose that we are fitting a straight line and wish to make standard error of the slope as small as possible. Suppose that the region of interest for x is $-1 \leq x \leq 1$. Where should the observations $x_1, x_2, \dots, x_n$ be taken?
5. Consider the simple linear regression model $y = \beta_0 + \beta_1 x + \varepsilon$, with usual assumptions on ε . Also assume that β_0 is known. Find the LSE of β_1 and its' variance.
6. Show that $Var(\hat{\mathbf{y}}) = \sigma^2 H$.
7. Consider the multiple linear regression model $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$. Show that LSE of $\boldsymbol{\beta}$ can be written as $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + R\boldsymbol{\varepsilon}$, where $R = (X^T X)^{-1} X^T$.
8. Prove that R^2 is the square of the sample correlation between $\mathbf{y}$ and $\hat{\mathbf{y}}$.
9. Consider the following two models with usual assumptions on error. Model A: $\mathbf{y} = X_1 \boldsymbol{\beta}_1 + \varepsilon$, Model B: $\mathbf{y} = X_1 \boldsymbol{\beta}_1 + X_2 \boldsymbol{\beta}_2 + \varepsilon$. Show that $R_A^2 \leq R_B^2$.
10. Consider the linear regression $y = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p + \epsilon$, with usual assumptions on ϵ . Assuming that X is a full column rank matrix, show that

$$\sum_{i=1}^n Var(\hat{y}_i) = (p+1)\sigma^2.$$

11. Show that $e_{(i)} = \frac{e_i}{1-h_{ii}}$.